THE PROBLEM · FRESHNESSGUARD AI

AI Answers Have an Expiry Date.

Keep the original answer. Verify the claims against live external evidence.
A citation-verified freshness layer powered by Bifrost + Tavily.



PRESERVE TEXT

Zero Silent Rewriting

Original generation left completely intact.

AUDITABLE VERDICTS

Evidence-Backed Signals

Deterministic passage matching via Tavily + Bifrost.

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THE TEMPORAL DRIFT TRAP

A Confident Answer Can Still Be Outdated.

Static weights fail when production reality moves faster than training checkpoints.

TEMPORAL DRIFT

Ground truth moves

LLMs draw on historical training weights. Patches, end-of-life notices and API changes can arrive weekly; pricing, availability, policies, product details and terms can change between training checkpoints.

FALSE SENSE OF SECURITY

Citations are not verification

A link may lead to a documentation root, an obsolete changelog or content contradicting the stated version or date. A citation alone does not establish support for the exact claim today.

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CORE ARCHITECTURAL PRINCIPLES

Verification Without Silent Rewriting.

Preserve the original answer. No qualifying evidence? No fabricated certainty.

THE VERIFICATION FLOW

01 Extract claims
02 Search for evidence
03 Validate citations
04 Report the verdict

The original answer stays intact; a separate report explains the result.

SAVED REPORT / CONCEPTUAL VIEW

Evidence before certainty

“The latest stable Python release series is 3.11.”

UNCERTAIN → OUTDATED

Captured live result → separate offline corrected-evidence replay. See slide 7.

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END-TO-END PIPELINE

From AI Answer to Evidence-Backed Verdict.

Distinguishing model judgment from deterministic string validation across six connected stages.

01 Input answer
Preserve original output intact.

02 Extract
Time-sensitive claims isolated.

03 Tavily search
External evidence harvested.

04 Rank & retrieve
Prioritize official docs.

05 Bifrost + Gate
Model judgment + deterministic validation.

06 Verdict report
Explain and cite evidence.

Deterministic safeguards: Check exact quotes, dates and versions; bound requests; handle uncertainty and rate limits.

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SYSTEM ARCHITECTURE

Two Technologies. One Verification Pipeline.

Coordinating model routing, live web retrieval, and deterministic passage verification.

MODEL GATEWAY **Bifrost**	Routes model calls for claim extraction and evidence judgment. Configured model routing
WEB RETRIEVAL **Tavily**	Searches the web for bounded evidence candidates. Targeted Tavily search
DETERMINISTIC CORE **FreshnessGuard**	Selects sources; validates exact quotes, citations and release evidence. Deterministic checks + abstention

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OFFLINE TEST SUITE & REGRESSION RIGOR

Built. Tested. Reproducible.

25/25

Scripted cases

Earlier scripted run passed

173

Python tests

Earlier local run: all passed

14

Frontend tests

Passed; production build passed

9/9

Adverse proposals

Earlier scripted rejections

Separate restricted-environment run (not combined with the results above):

Python: 173 discovered, 166 passed, 1 failed, 6 errors. Frontend: 14 executed, 8 passed, 6 failed.

Local socket and environment restrictions affected this execution.

Previously reported local test results; a separate restricted-environment run encountered socket and environment-related failures. Scripted results are not measured live accuracy.

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When Evidence Falls Short, We Say So.

A. NOV 10 — LIVE VERIFICATION

OUTDATED

0% Freshness · 100% Evidence Coverage

1 accepted official Python citation

"The latest stable Python release series is 3.11."

Evidence: Official Python 3.14.0 stable release page, released October 7, 2025.

Genuine live session through the configured Bifrost gateway, Gemini model integration, Tavily evidence retrieval, and FreshnessGuard's deterministic validation pipeline.

B. NOV 6 — EARLIER LIVE VERIFICATION

UNCERTAIN

The proposed quotation omitted the release date required by the evidence validator.

No accepted citation; the report stayed Uncertain.

C. OFFLINE CORRECTED-EVIDENCE REPLAY

OUTDATED

A complete dated passage produced an Outdated verdict using saved evidence.

No new provider or search calls.

Evidence-backed when decisive. Explicitly uncertain when not.

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INSPECT & REPRODUCE

AI Answers Deserve Verifiable Evidence.

Preserve the answer. Verify what changes. Make uncertainty visible.

FRESHNESSGUARD AI

Bifrost + Tavily + evidence you can inspect

GitHub Repository

Watch Live Demo

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